

Lab 13 Activity

Part 1: Logistic Regression

1

Load in the MplsStops data from the carData package.

```
data(MplsStops, package = "carData")
summary(MplsStops)
```

```
      idNum      date      problem
17-000003: 1  Min.    :2017-01-01 00:00:42  suspicious:25822
17-000007: 1 1st Qu.:2017-03-29 08:35:09  traffic    :26098
17-000073: 1  Median :2017-06-17 18:46:47
17-000092: 1  Mean   :2017-06-23 19:57:49
17-000098: 1 3rd Qu.:2017-09-18 18:32:06
17-000111: 1  Max.    :2017-12-31 23:52:35
(Other)   :51914
```

```
      MDC      citationIssued personSearch vehicleSearch
MDC   :43699  NO :15899      NO :38462    NO :40579
other: 8221  YES : 3211      YES : 5237    YES : 3120
      NA's:32810  NA's: 8221  NA's: 8221
```

```
      preRace      race      gender      lat
Unknown      :28337  Black      :15220  Female :10015  Min.    :44.89
Black        : 6805  White      :11703  Male   :27131  1st Qu.:44.95
White        : 6004  Unknown    : 9219  Unknown: 6492  Median :44.98
Native American: 908  East African: 2188  NA's   : 8282  Mean   :44.97
Latino       :  528  Latino     : 1858                3rd Qu.:45.00
```

(Other)	: 1117	(Other)	: 3511	Max.	:45.05
NA's	: 8221	NA's	: 8221		
long	policePrecinct	neighborhood			
Min.	:-93.33	Min.	:1.000	Downtown West:	4409
1st Qu.	:-93.29	1st Qu.	:2.000	Whittier	: 3328
Median	:-93.28	Median	:3.000	Near - North	: 2256
Mean	:-93.27	Mean	:3.257	Lyndale	: 2154
3rd Qu.	:-93.25	3rd Qu.	:4.000	Jordan	: 2075
Max.	:-93.20	Max.	:5.000	Hawthorne	: 2031
			(Other)	:35667	

Look at the documentation to understand the context. Based on this information alone, select 2 or 3 predictors that you think might predict whether a citation was issued.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
data(MplsStops, package = "carData")
```

```
summary(MplsStops)
```

idNum		date	problem
17-000003:	1	Min. :2017-01-01 00:00:42	suspicious:25822
17-000007:	1	1st Qu.:2017-03-29 08:35:09	traffic :26098
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17-000092:	1	Mean :2017-06-23 19:57:49	
17-000098:	1	3rd Qu.:2017-09-18 18:32:06	
17-000111:	1	Max. :2017-12-31 23:52:35	
(Other)	:51914		
MDC		citationIssued	personSearch vehicleSearch
MDC :43699	NO :15899	NO :38462	NO :40579
other: 8221	YES : 3211	YES : 5237	YES : 3120

NA's:32810 NA's: 8221 NA's: 8221

	preRace		race		gender		lat
Unknown	:28337	Black	:15220	Female	:10015	Min.	:44.89
Black	: 6805	White	:11703	Male	:27131	1st Qu.:	:44.95
White	: 6004	Unknown	: 9219	Unknown:	6492	Median	:44.98
Native American:	908	East African:	2188	NA's	: 8282	Mean	:44.97
Latino	: 528	Latino	: 1858			3rd Qu.:	:45.00
(Other)	: 1117	(Other)	: 3511			Max.	:45.05
NA's	: 8221	NA's	: 8221				
	long		policePrecinct		neighborhood		
Min.	:-93.33	Min.	:1.000	Downtown West:	4409		
1st Qu.:	-93.29	1st Qu.:	2.000	Whittier	: 3328		
Median	:-93.28	Median	:3.000	Near - North	: 2256		
Mean	:-93.27	Mean	:3.257	Lyndale	: 2154		
3rd Qu.:	-93.25	3rd Qu.:	4.000	Jordan	: 2075		
Max.	:-93.20	Max.	:5.000	Hawthorne	: 2031		
				(Other)	:35667		

```
# choice of variables is open - nearly any choice will be categorical
# race, gender, and policePrecinct will be used below
# note that policePrecinct should be converted to factor if used

MplsStops <- MplsStops %>%
  mutate(policePrecinct = factor(policePrecinct))
```

2

Not all rows have information about whether a citation was issued. Remove these rows. Then, use `mutate` to transform `citationIssued` into a numeric variable of 0's (no citation) and 1's (yes citation).

```
library(tidyverse)
MplsStops <- MplsStops %>%
  filter(!is.na(citationIssued)) %>% #this removes missing values
  mutate(citationIssued0 = as.numeric(citationIssued) - 1)

# check that the variable works
table(MplsStops$citationIssued, MplsStops$citationIssued0)
```

	0	1
NO	15899	0
YES	0	3211

3

Try out a series of models using the variables that you identified in (1). For learning's sake, you may first remove rows with missing values on the predictors, but you might also try to figure out how missing data methods might apply to logistic regression. Use classification accuracy and model comparisons to select the best-performing model. What do you find?

```
MplsStops <- MplsStops %>% #next line removes rows with missing values on predictors
  filter(!is.na(gender) & !is.na(race) & !is.na(policePrecinct))

mod1 <- glm(citationIssued0 ~ gender + race + policePrecinct, MplsStops,
            family = binomial)
fitted1 <- fitted(mod1)
class1 <- ifelse(fitted1 > .5, "pred_1", "pred_0")
table(class1, MplsStops$citationIssued0)
```

class1	0	1
pred_0	15874	3209
pred_1	1	0

```
# everyone is predicted to not have a citation - maybe if we use a different cutoff

class1 <- ifelse(fitted1 > .2, "pred_1", "pred_0")
table(class1, MplsStops$citationIssued0)
```

class1	0	1
pred_0	10319	1322
pred_1	5556	1887

```
(5173 + 929) / (5173 + 673 + 2767 + 929) # 64% accuracy
```

```
[1] 0.6394886
```

```
summary(mod1)
```

Call:

```
glm(formula = citationIssued0 ~ gender + race + policePrecinct,  
     family = binomial, data = MplsStops)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.13708	0.06781	-16.770	< 2e-16	***
genderMale	-0.06223	0.04699	-1.324	0.185362	
genderUnknown	0.41431	0.13173	3.145	0.001661	**
raceWhite	-0.26459	0.05111	-5.176	2.26e-07	***
raceUnknown	-1.98252	0.11874	-16.696	< 2e-16	***
raceEast African	-0.32482	0.09993	-3.251	0.001152	**
raceLatino	0.28974	0.09092	3.187	0.001438	**
raceNative American	-0.56161	0.12215	-4.598	4.27e-06	***
raceOther	-0.64587	0.12822	-5.037	4.72e-07	***
raceAsian	-0.07307	0.15433	-0.473	0.635872	
policePrecinct2	0.47516	0.06686	7.106	1.19e-12	***
policePrecinct3	-0.42992	0.07561	-5.686	1.30e-08	***
policePrecinct4	-0.07755	0.06543	-1.185	0.235932	
policePrecinct5	-0.26693	0.06926	-3.854	0.000116	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 17288 on 19083 degrees of freedom
Residual deviance: 16179 on 19070 degrees of freedom
AIC: 16207

Number of Fisher Scoring iterations: 5

```
# mod2: remove gender  
mod2 <- glm(citationIssued0 ~ race + policePrecinct, MplsStops,  
            family = binomial)  
fitted2 <- fitted(mod2)  
class2 <- ifelse(fitted2 > .2, "pred_1", "pred_0")  
table(class2, MplsStops$citationIssued0)
```

```
class2      0      1
  pred_0 10350  1325
  pred_1  5525  1884
```

```
(5298 + 891) / (5298 + 711 + 2642 + 891) # 65% accuracy - a bit better
```

```
[1] 0.6486062
```

```
anova(mod1, mod2, test = "LRT") # the more complicated model fits significantly better, but v
```

Analysis of Deviance Table

Model 1: citationIssued0 ~ gender + race + policePrecinct

Model 2: citationIssued0 ~ race + policePrecinct

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	19070	16179			
2	19072	16194	-2	-15.66	0.0003975 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# mod3: use only race
mod3 <- glm(citationIssued0 ~ race, MplsStops,
            family = binomial)
fitted3 <- fitted(mod3)
class3 <- ifelse(fitted3 > .2, "pred_1", "pred_0")
table(class3, MplsStops$citationIssued0)
```

```
class3      0      1
  pred_0  9973  1447
  pred_1  5902  1762
```

```
(5029 + 882) / (5029 + 720 + 2911 + 882) # 62% accuracy = a bit worse
```

```
[1] 0.6194718
```

```
anova(mod2, mod3, test = "LRT") # mod that includes policePredict fits significantly better
```

Analysis of Deviance Table

Model 1: citationIssued0 ~ race + policePrecinct

Model 2: citationIssued0 ~ race

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	19072	16194			
2	19076	16424	-4	-229.78	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# In this example, I'll retain mod2, but arguments can be made for other conclusions
```

Part 2: Multilevel Regression

For this part, we'll work with a dataset that contains weight measurements on multiple occasions for Asian children born in Britain. Load in the necessary packages and the dataset using the following code:

```
library(tidyverse)
library(haven)
library(lme4)
dat <- read_dta("Lab_13_asian.dta")
dat %>% glimpse #already in long format
```

Rows: 198

Columns: 6

```
$ id      <dbl> 45, 45, 45, 45, 45, 258, 258, 258, 258, 287, 287, 287, 287, 483~
$ occ     <dbl> 1, 2, 3, 4, 5, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 1, 2, 1, 2, 1, ~
$ age     <dbl> 0.1368925, 0.6570842, 1.2183436, 1.4291581, 2.2724161, 0.191649~
$ weight  <dbl> 5.171, 10.860, 13.150, 13.200, 15.880, 5.300, 9.740, 9.980, 11.~
$ brthwt  <dbl> 4140, 4140, 4140, 4140, 4140, 3155, 3155, 3155, 3155, 3850, 385~
$ gender  <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, ~
```

- id: child identifier
- occ: numbered occasion of each measurement
- age: child's age (in years) at each measurement
- weight: child's measured weight (kg)
- brthwt: child's birthweight
- gender: 1 = male, 2 = female

4

Fit a simple (no covariates) variance components model predicting child's weight using these data. What problem do you run into? Considering the nature of these data, what do you think might be the problem?

```
library(lme4)
summary(lmer(weight ~ (1 | id), dat))
```

boundary (singular) fit: see help('isSingular')

Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ (1 | id)
Data: dat

REML criterion at convergence: 1000.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.89516	-0.60438	0.07727	0.67815	2.76256

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.000	0.000
Residual		9.164	3.027

Number of obs: 198, groups: id, 68

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	8.8371	0.2151	41.08

optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```
## model is singular - notice that there is no variance associated with id
## children will span about the same range of weights over time - difficult to
## predict weight without including age as a covariate
```

5

Add one covariate to the model that you think will lead to the highest residual ICC. Calculate the residual ICC.


```
## age probably makes the most sense
## an argument could be made for including occasion, but age should have a more
## direct relationship with weight than occasion
## gender and birthwt probably aren't major factors
mod <- lmer(weight ~ age + (1 | id), dat)
summary(mod)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ age + (1 | id)
Data: dat
```

```
REML criterion at convergence: 701.4
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-2.06569	-0.67935	-0.01896	0.68187	2.41589

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.622	0.7887
	Residual	1.535	1.2390

Number of obs: 198, groups: id, 68

```
Fixed effects:
```

	Estimate	Std. Error	t value
(Intercept)	5.1566	0.1812	28.45
age	3.3895	0.1156	29.33

```
Correlation of Fixed Effects:
```

```
(Intr)
age -0.686
```

```
.622 / (.622 + 1.535) #.288
```

```
[1] 0.2883635
```

6

Given the variables available in this dataset, what combination of variables do you think might give the most realistic and accurate model of weight? Be creative, and remember what you learned about regression - many of the same principles apply! What ICC do you get?

```
cor(dat)
```

```
      id      occ      age      weight      brthwt
id      1.00000000 -0.02035928 -0.044373836 -0.1218791 -0.217677618
occ     -0.02035928  1.00000000  0.754329227  0.7274967  0.039087646
age     -0.04437384  0.75432923  1.000000000  0.8746259 -0.001733001
weight -0.12187911  0.72749668  0.874625915  1.0000000  0.170234021
brthwt -0.21767762  0.03908765 -0.001733001  0.1702340  1.000000000
gender  0.16568665 -0.01854712  0.005959521 -0.1161066 -0.064465841
      gender
id      0.165686654
occ     -0.018547116
age      0.005959521
weight -0.116106633
brthwt -0.064465841
gender  1.000000000
```

```
## probably shouldn't use both age and occasion since these are highly correlated

## gender probably should be included in the model

## age might have a nonlinear effect! could include age^2 as a covariate

## birthweight as a level-2 covariate could also make sense -
## but using this will lead lmer to give a warning about the scale of this variable

summary(lmer(weight ~ age + I(age^2) + gender + brthwt + (1 | id), dat))
```

Warning: Some predictor variables are on very different scales: consider rescaling

Linear mixed model fit by REML ['lmerMod']

Formula: weight ~ age + I(age^2) + gender + brthwt + (1 | id)

Data: dat

REML criterion at convergence: 547.3

Scaled residuals:

```
      Min      1Q  Median      3Q      Max
-2.2257 -0.5324 -0.0013  0.4769  3.9030
```

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	0.4717	0.6868
Residual		0.5505	0.7420

Number of obs: 198, groups: id, 68

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	1.277808	0.724831	1.763
age	7.832863	0.288139	27.184
I(age^2)	-1.714732	0.107996	-15.878
gender	-0.630511	0.200685	-3.142
brthwt	0.001001	0.000196	5.107

Correlation of Fixed Effects:

	(Intr) age	I(g^2) gender
age	-0.193	
I(age^2)	0.177	-0.970
gender	-0.498	-0.006
brthwt	-0.883	0.031

fit warnings:

Some predictor variables are on very different scales: consider rescaling

```
.4717 / (.4717 + .5505)
```

```
[1] 0.4614557
```

```
## this data set is analyzed in Multilevel and Longitudinal Modeling Using Stata  
## by Rabe-Hesketh and Skrondal - they use age, age^2, and gender as covariates  
## (along with a random slope for age, which we don't cover in this course)
```